

SOUTH BEND, IN, USA

WMO: 725350

Lat: 41.707N

Lon: 86.316W

Elev: 773

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14848

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.4	(c) 5.0	(d) -7.4	(e) 3.8	(f) 0.3	(g) -2.0	(h) 5.1	(i) 6.3	(j) 26.4	(k) 28.3	(l) 24.4	(m) 28.6	(n) 9.7	(o) 230	(p) 0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	90.1	73.9	87.5	72.4	84.9	71.2	77.2	85.9	75.3	83.5	73.7	81.4	11.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	132.3	82.6	72.6	124.5	80.2	71.1	117.9	78.6	41.0	85.8	39.2	83.8	37.7	81.4	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.1	(b) 20.6	(c) 18.6	(e) -7.9	(f) 94.6	(g) 7.1	(h) 3.0	(i) -13.0	(j) 96.7	(k) -17.1	(l) 98.5	(m) -21.1	(n) 100.2	(o) -26.3	(p) 102.3		
(4) 24.1	(5) 20.6	(6) 18.6	(7) -7.9	(8) 94.6	(9) 7.1	(10) 3.0	(11) -13.0	(12) 96.7	(13) -17.1	(14) 98.5	(15) -21.1	(16) 100.2	(17) -26.3	(18) 102.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	24.5	27.8	37.5	49.2	59.8	69.6	73.0	71.4	64.9	53.0	40.5	30.4
	DBStd	19.29	11.67	10.98	11.28	9.67	9.01	7.10	5.74	5.30	7.60	8.79	9.52	10.33
	HDD50	2975	790	624	418	131	17	0	0	0	1	68	313	613
	HDD65	6187	1254	1041	855	483	214	35	6	10	95	384	736	1074
	CDD50	3068	1	2	31	105	321	588	714	663	449	162	27	5
	CDD65	804	0	0	2	7	54	173	255	208	92	13	0	0
	CDH74	6856	0	0	16	74	524	1537	2257	1593	747	108	0	0
	CDH80	2086	0	0	2	11	140	511	765	429	211	17	0	0
Wind	WSAvg	8.9	10.6	10.3	10.2	10.6	9.0	7.7	7.0	6.5	7.0	8.6	9.6	9.9
Precipitation	PrecAvg	39.1	2.4	2.2	2.7	3.6	3.6	4.1	3.8	3.9	3.5	3.3	3.0	2.8
	PrecMax	55.6	5.4	8.1	8.0	7.5	8.3	10.9	8.7	12.8	13.9	9.1	6.7	5.5
	PrecMin	25.2	0.7	0.6	0.8	0.5	0.7	0.5	1.2	0.4	0.0	0.9	0.3	0.9
	PrecStd	6.1	1.2	1.3	1.4	1.5	1.9	2.2	1.7	2.4	2.4	1.9	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	61.9	76.2	81.3	88.3	92.7	94.8	90.8	90.1	82.3	69.2	61.1
		MCWB	53.7	54.1	62.5	63.3	68.8	73.7	78.0	76.3	72.0	67.6	56.6	57.5
	2%	DB	51.0	54.5	68.1	75.4	83.9	89.1	90.2	87.6	85.8	77.0	63.4	54.9
		MCWB	48.4	49.0	57.2	60.3	67.4	71.8	75.4	73.9	70.2	64.2	56.3	51.0
	5%	DB	44.8	47.9	61.2	71.0	80.1	85.8	87.3	84.6	81.8	72.2	59.5	48.8
		MCWB	41.7	43.3	53.2	57.8	65.7	70.9	73.4	72.3	68.7	62.1	53.5	44.9
	10%	DB	39.6	42.5	55.2	66.1	75.9	82.5	84.2	82.0	78.1	67.8	55.2	43.7
		MCWB	37.1	38.5	48.3	54.6	63.5	69.4	71.9	70.8	66.4	59.8	49.4	40.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	56.2	63.6	67.2	73.2	77.1	80.3	78.7	75.4	70.9	61.4	58.5
		MCDB	57.1	59.6	76.0	77.6	82.9	88.0	91.0	87.9	85.3	79.0	65.6	61.2
	2%	WB	48.5	49.4	58.9	62.8	70.2	75.0	78.0	76.4	73.0	67.2	57.9	51.6
		MCDB	50.3	53.5	66.0	71.3	79.5	85.0	86.8	84.1	81.2	72.9	61.7	54.1
	5%	WB	41.8	43.4	54.0	59.4	68.1	73.2	75.9	74.6	70.9	63.8	53.9	45.6
		MCDB	44.1	47.7	60.0	68.3	76.9	82.2	84.1	81.4	77.8	70.0	58.4	48.4
	10%	WB	37.2	38.6	48.6	56.3	65.5	71.3	74.1	72.9	68.8	60.5	50.1	40.6
		MCDB	39.6	42.1	54.8	65.2	73.2	79.6	81.6	79.3	75.6	67.0	54.9	43.4
Mean Daily Temperature Range	5% DB	MDBR	13.1	14.3	17.4	20.4	20.2	19.2	18.8	18.2	19.7	18.0	14.5	12.0
		MCDBR	17.0	20.7	24.5	25.7	23.9	21.7	20.9	20.3	22.4	23.6	19.4	17.2
		MCWBR	14.8	16.6	16.4	14.8	12.2	9.6	9.3	9.3	10.1	13.0	14.0	14.9
	5% WB	MCDBR	16.2	19.3	21.8	22.9	21.0	18.7	18.4	16.8	18.8	19.9	17.1	15.9
		MCWBR	15.0	16.6	15.8	15.0	12.5	9.7	9.7	8.9	9.8	13.3	14.3	14.9
Clear-Sky Solar Irradiance	taub	0.301	0.325	0.345	0.395	0.424	0.430	0.442	0.438	0.414	0.355	0.318	0.299	
	taud	2.392	2.330	2.366	2.244	2.222	2.248	2.213	2.218	2.260	2.429	2.479	2.449	
	Ebn at Noon	268	278	286	277	271	269	264	261	259	265	261	260	
	Edh at Noon	25	32	35	42	44	43	44	43	38	29	23	23	
All-Sky Solar Radiation	RadAvg	462	724	1124	1479	1690	1927	1948	1689	1385	897	552	388	
	RadStd	44	95	95	128	155	121	99	98	115	95	79	42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page